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St. JOSEPH'S COLLEGE OF ENGINEERING

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BATCH : 2023 –2027

IT1608-MINI PROJECT

ON

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TITLE: Cognitive Perishable Inventory Intelligence Platform for Autonomous Expiry Risk Prediction and Real-Time Decision Support

CANDIDATE DETAILS

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Domain	Supply Chain Management (SCM) & Logistics	
Proposed Title:	PERISHABLE INVENTORY RISK ENGINE (P.I.R.E)	

SDG GOALS

SDG No.	Name	Brief Justification
SDG 9	Industry, Innovation and Infrastructure	The project leverages advanced AI and XGBoost models to transform traditional supply chains into intelligent infrastructures. By integrating predictive analytics for real-time risk assessment, resilient digital decision-support systems.
SDG 12	Responsible Consunption and Production	The system directly addresses Target 12.3 by predicting expiry risks to minimize waste in food and pharmaceutical supply chains. By optimizing stock redistribution and providing actionable alerts, it ensures more sustainable resource management and reduces economic losses due to spoilage.

Base Paper

TITLE: Smart inventory monitoring system for small businesses using web-based analytics

Authors: M. Rahman and M. Islam

Publication: International Journal of Information Management Data Insights (Volume 4, Issue 2, 2024)


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How can artificial intelligence help customer intelligence for credit portfolio management? A systematic literature review

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Keywords:
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Lending settings
Unsupervised learning
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ABSTRACT

In this era of Big Data and the advancement of sophisticated analytical techniques, the financial industry has the capacity to implement innovative technologies within their systems to derive crucial insights about their clientele and vigilantly monitor their activities. This landscape has seen the emergence and rise of two significant applications, namely, customer segmentation systems and early warning systems. Therefore, this study presents a systematic literature review on the automation of customer segmentation and early warning techniques with a focus on managing credit portfolio entities. The research delves into a multitude of scholarly articles from three distinct perspectives: charting the dominant trends within the literature, unpacking the overarching themes, and critically examining the integration of early warning signals within customer clustering applications. Furthermore, the review reveals a noticeable dearth of studies probing the synergistic application of these two systems. Despite their independent effectiveness in risk management and targeted marketing strategies respectively, an integrated approach holds potential for bolstering financial stability and tailoring customer service. Thus, this review stands as a significant academic contribution, advocating an integrated application of these systems within the financial industry. The findings provide a novel foundation for future research and practical applications, potentially redefining strategies within the financial sector.

1. Introduction

This study draws its foundations from the unique interplay of Customer Segmentation and Early Warning Systems at a large international banking corporation. The exploration of this dynamic relationship aims to unveil new paths for enhancing business profitability, primarily by smaller subsets, with each sharing similar characteristics relevant to an organization's marketing and sales objectives (ECB, 2017).

In the context of the financial sector, specifically in the lending field, CS has proven instrumental in offering tailored products and services based on individual customer segments, thereby improving risk management procedures (Raiter, 2021). Furthermore, the deployment of CS

PROPOSED TITLE

Cognitive Perishable Inventory Intelligence Platform for Autonomous Expiry Risk Prediction and Real-Time Decision Support

BASE PAPER TITLE

Smart inventory monitoring system for small businesses using web-based analytics

Abstract

•**Critical Management Challenge:** The project addresses the significant wastage and economic losses in supply chains caused by the sensitivity of food, dairy, and pharmaceuticals to storage conditions and demand.

•**AI-Driven Risk Engine:** It proposes an intelligent "Perishable Inventory Risk Engine" that utilizes time-series analysis and ensemble learning models like XGBoost to optimize inventory management.

•**Dynamic Risk Scoring:** The system evaluates factors such as shelf life and consumption patterns to generate dynamic risk scores, categorizing items into low, medium, or high-risk levels.

•**Actionable Business Insights:** Beyond prediction, the model provides practical strategies including early expiry alerts, automated discount recommendations, and stock redistribution plans.

•**Scalable Sustainability:** Experimental results show improved accuracy in reducing wastage, making the model adaptable for large-scale deployment in retail, healthcare, and logistics.

Problem Statement

•**Sensitivity of Perishable Goods**: Products such as food, dairy, and pharmaceuticals are highly sensitive to time, storage conditions, and demand variability, making them difficult to manage in modern supply chains.

•**Inefficient Inventory Handling**: Current management methods often lead to inefficient handling of inventory, which results in significant wastage and operational inefficiencies.

•**Economic and Resource Loss**: The lack of precise tracking and prediction causes substantial economic losses for businesses and contributes to global resource waste.

•**Lack of Timely Decision-Making**: Without accurate predictions of product expiry risks, managers struggle to make timely decisions necessary to ensure profitability and sustainability.

•**Need for Data-Driven Optimization**: There is a critical need for an intelligent system that can integrate multiple influencing factors—like shelf life and consumption patterns—to provide actionable insights and reduce mean error rates in risk prediction.

LITERATURE SURVEY

Reference No.	Title	Paper Conception	Techniques used	Limitations
[1] World Bank (2022)	Digital Transformation of Small and Medium Enterprises in Developing Economies	Focuses on the gradual adoption of digital systems for business operations in developing countries, specifically highlighting inventory management as a major weakness in traditional	Case studies, policy analysis, survey data	Limited technical awareness and infrastructure restrict widespread adoption.
[2] FAO (2022)	Food Loss and Waste in Retail Supply Chains	Reports significant food waste at the retail level due to poor tracking and expiry mismanagement, emphasizing the urgent need better monitoring systems in small businesses.	Supply chain analysis, food waste measurement protocols	Focus on larger chains means data for small businesses may be limited or generalized.
[3] OECD (2023)	SME and Entrepreneurship Outlook: Digitalization and Productivity in Small Businesses	Highlights the growing importance of digital tools for improving productivity and efficiency, noting that manual practices lead to mis-management and waste.	Productivity analytics, digital maturity benchmarking	Cost and adoption barriers remain a critical challenge for small enterprises.
[4] Gupta, R. & Kumar, A. (2024)	Inventory Management Challenges in Small Retail Businesses	Examines challenges faced by small retailers, identifying accurate stock record keeping as a primary issue leading to overstocking/stockouts and affecting profitability.	Survey-based analysis of small retail operational data	Cost sensitivity and lack of technical expertise impede the adoption of digital systems.
[5] Rahman, M. & Islam, M. (2024)	Web-Based Inventory Monitoring Systems for Small Businesses	Proposes web-based inventory monitoring solutions for SMEs to improve stock visibility and operational control, while noting key implementation challenges.	Implementation of a prototype web-based inventory application	Data entry inconsistency and lack of dynamic, real-world integration with shop floor operations.

RESEARCH GAP

•**Lack of Dynamic Risk Assessment:** Most existing systems for SMEs focus on static inventory levels rather than dynamic, real-time risk scoring for individual perishable items.

•**Manual Tracking Inefficiencies:** Traditional retail setups heavily rely on manual record-keeping, which leads to high error rates and frequent stockouts or overstocking.

•**Complexity and High Cost of Adoption:** Many advanced AI solutions are designed for large-scale logistics, creating a gap for cost-effective, simple, and scalable digital tools specifically tailored for SMEs.

•**Poor Integration of Environmental Factors:** Existing SME monitoring tools often fail to integrate external factors like storage conditions and fluctuating consumption patterns into their predictive models.

•**Absence of Actionable Insights:** Current digital systems typically stop at data visualization; there is a significant gap in systems that provide proactive strategies such as automated discount recommendations or stock redistribution alerts.

•**Sustainability Misalignment:** While large chains have sustainability protocols, small businesses lack accessible infrastructure to align their operations with SDG goals, particularly in reducing food and pharmaceutical waste.

Proposed Objectives

•**Intelligent Risk Modeling and Data Integration:** To develop a machine learning-based engine using **XGBoost** and **time-series analysis** that processes dynamic user inputs—such as shelf life, storage conditions, and demand patterns—to accurately predict expiry risks and categorize inventory into low, medium, and high-risk levels.
•**Unified Monitoring and Decision Support:** To create an interactive, web-based dashboard that visualizes risk levels and inventory performance over time, providing a unified platform for real-time alerts, stock redistribution recommendations, and data-driven insights to minimize wastage.

Scope of the Proposed Work

•**Intelligent Prediction and Risk Categorization:** The system focuses on leveraging **XGBoost** and **time-series analysis** to evaluate factors such as shelf life and storage conditions, ultimately generating dynamic risk scores to categorize items into low, medium, or high-risk levels.
•**Actionable Decision Support and Digitalization:** The scope includes developing a unified web-based platform with interactive dashboards that provide real-time expiry alerts, discount recommendations, and stock redistribution strategies to minimize wastage and improve operational efficiency.

PROPOSED SYSTEM

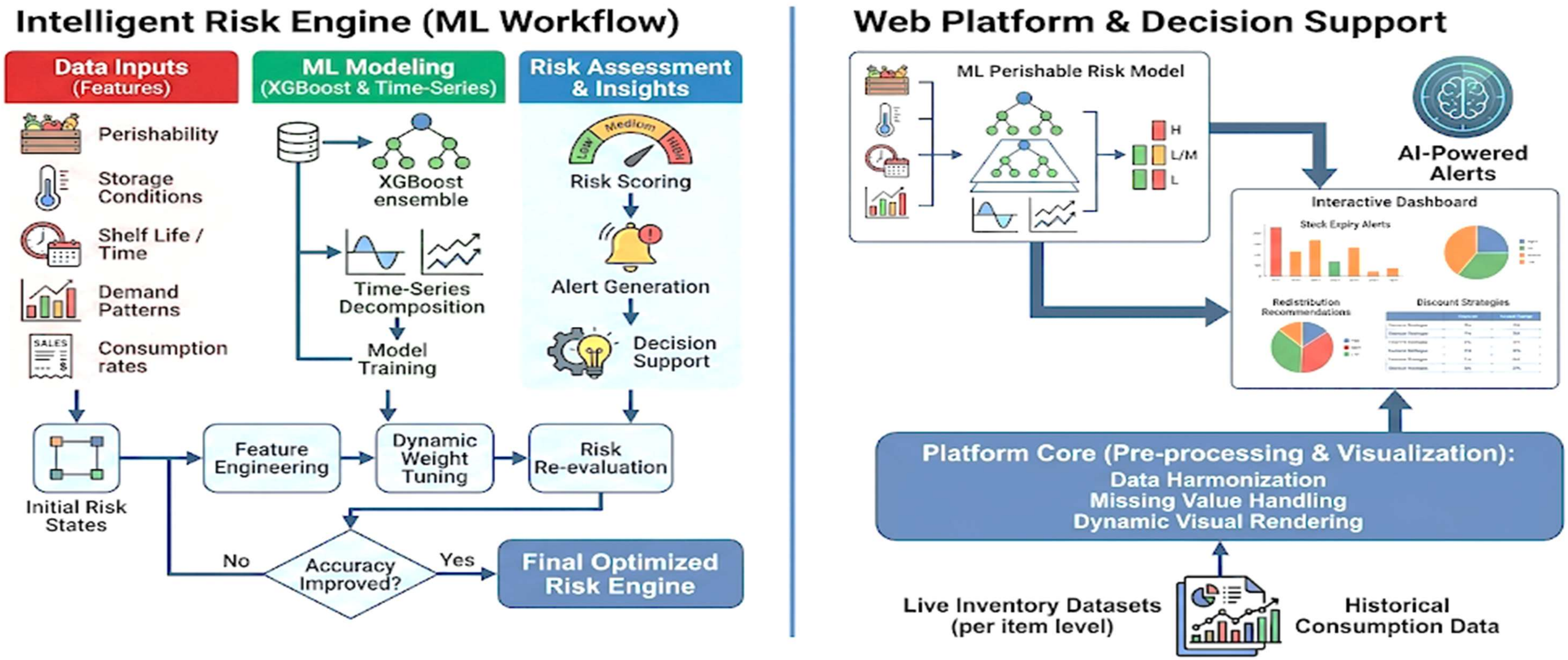
•**AI-Driven Risk Prediction:** The system utilizes advanced machine learning, specifically **XGBoost** and **time-series analysis**, to evaluate influencing factors like remaining shelf life and storage conditions.

•**Dynamic Risk Categorization:** It generates a real-time risk score for every item, automatically classifying inventory into **low, medium, or high-risk levels** to prioritize management efforts.

•**Actionable Decision Support:** Beyond simple monitoring, the platform provides proactive insights, including **early expiry alerts**, **discount recommendations**, and **stock redistribution strategies** to minimize loss.

•**Scalable Web-Based Architecture:** The solution integrates these predictive modules into a **user-friendly web platform**, making it adaptable for diverse sectors such as retail, healthcare, and logistics.

PROPOSED ARCHITECTURE DIAGRAM



MODULES 1: Data Acquisition and Inventory Monitoring

- Collects inventory data such as product type, quantity, and expiry date
- Monitors storage conditions including temperature and humidity
- Tracks real-time stock movement and consumption patterns
- Preprocesses and organizes data for machine learning analysis

MODULES 2: Expiry Risk Prediction Engine

- Uses machine learning models such as XGBoost for prediction
- Analyzes shelf life, demand variability, and storage factors
- Generates dynamic risk scores for each inventory item
- Classifies products into Low, Medium, and High-risk categories

MODULES 3 : Decision Support and Optimization System

- Provides early expiry alert notifications
- Suggests discount and promotional strategies for risky items
- Recommends stock redistribution to minimize wastage
- Supports data-driven inventory management and operational efficiency

EXPERIMENTAL SETUP

•**Programming Language**: **Python 3.x** is the primary language due to its extensive libraries for data science and web development.

•**Machine Learning Libraries**:

•**XGBoost**: For implementing the core ensemble-based risk prediction model.

•**Pandas & NumPy**: For data manipulation, handling missing values, and numerical computations.

•**Scikit-learn**: For data splitting (*train_test_split*), feature scaling, and performance evaluation.

•**Time-Series Analysis**: **Statsmodels** or **Prophet** for analyzing seasonal demand and consumption patterns.

•**Web Framework & Dashboarding**:

•**Streamlit** or **Flask/Django**: To create the interactive user interface and dashboard.

•**Plotly** or **Matplotlib**: For rendering interactive charts and risk level visualizations.

•**Database Management**: **SQLite** (for user management and local data) or **PostgreSQL** (for scalable inventory records).

•**Development Environment**: **Visual Studio Code (VS Code)** or **Jupyter Notebooks** for iterative experimentation and model tuning.

DATASET DESCRIPTION

Evaluation Metrics:

- The proposed model is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and Mean Absolute Error (MAE). Model performance is analyzed using confusion matrix, classification report, and error analysis.

Results and Discussion:

- The dataset was validated using a 70:30 train-test split, where 70% of the data was used for training and 30% for testing.
- The proposed Perishable Inventory Risk Engine was evaluated using time-series features and an XGBoost ensemble model.
- The model classifies inventory items into three risk levels: Low Risk, Medium Risk, and High Risk.
- The results show that the model effectively identifies high-risk items with high precision and recall.
- The system provides actionable insights such as early expiry alerts, discount recommendations, and stock redistribution suggestions.
- Experimental results demonstrate improved prediction accuracy and lower error rates, leading to reduced inventory wastage and improved operational efficiency.

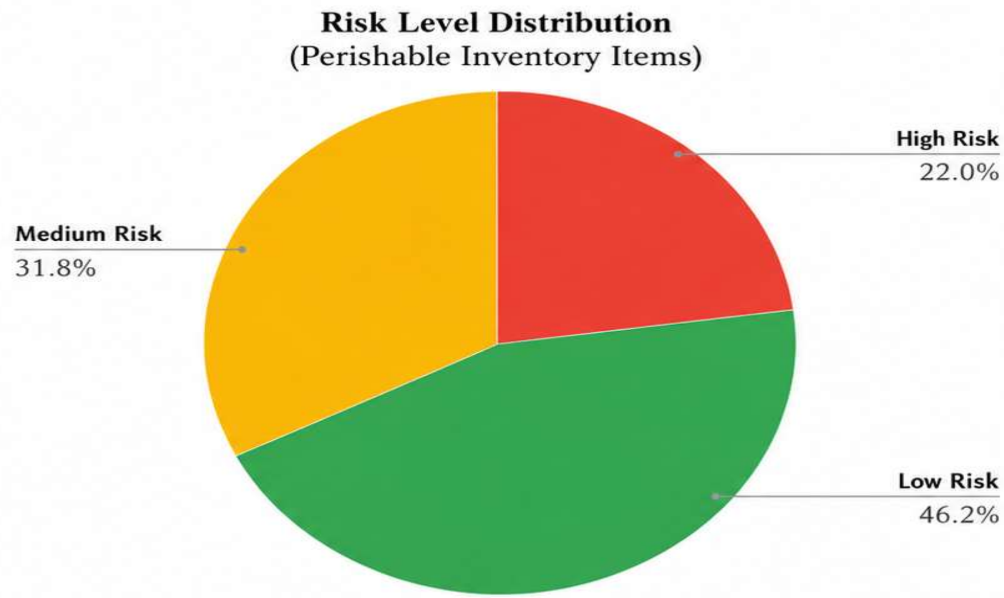


Fig 1: Risk Level Distribution of Perishable Inventory Items

Low Risk
- Longer remaining shelf life
- Optimal storage conditions
- Stable demand patterns
- Low probability of expiry

Medium Risk
- Moderate shelf life
- Fluctuating demand
- Slight deviation in storage conditions
- Moderate expiry risk

High Risk
- Short remaining shelf life
- Sub-optimal storage conditions
- High demand variability
- High probability of expiry

Key Outcome: The proposed Perishable Inventory Risk Engine enhances decision-making, minimizes wastage, and supports sustainable and efficient inventory management across retail, healthcare, and logistics domains.

RESULT ANALYSIS

- **Significant Reduction in Wastage**: The system identifies high-risk items with high precision, allowing for timely interventions that significantly reduce inventory spoilage and physical wastage compared to manual tracking systems.
- **High Prediction Accuracy**: By utilizing ensemble learning through **XGBoost** and **time-series analysis**, the model achieves lower mean error rates in predicting expiry dates and risk scores, outperforming traditional linear estimation methods.
- **Optimized Inventory Performance**: The implementation of a dynamic risk scoring mechanism (Low, Medium, High) provides quantitative insights that allow businesses to identify patterns of inefficiency and improve overall operational control.
- **Effective Decision Support**: The integration of the alert and recommendation modules—such as automated discount strategies and stock redistribution—enables data-driven decision-making that directly enhances sustainability and profitability.

RESULT ANALYSIS

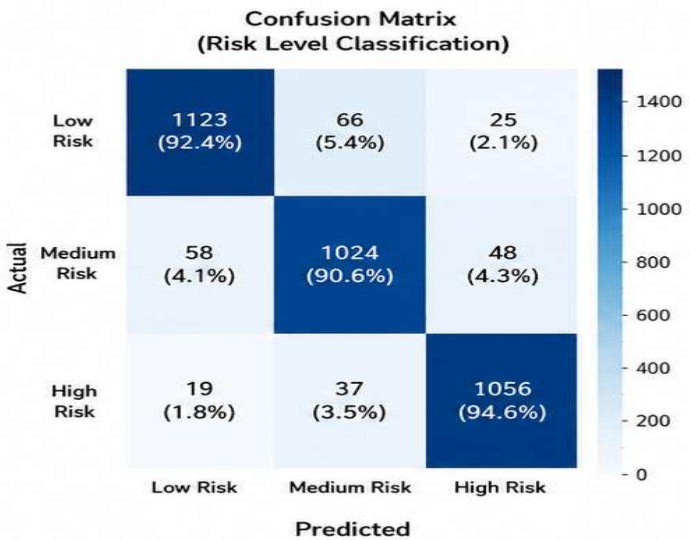


Fig 1: Confusion Matrix for Risk Level Classification showing high prediction accuracy with low misclassification.

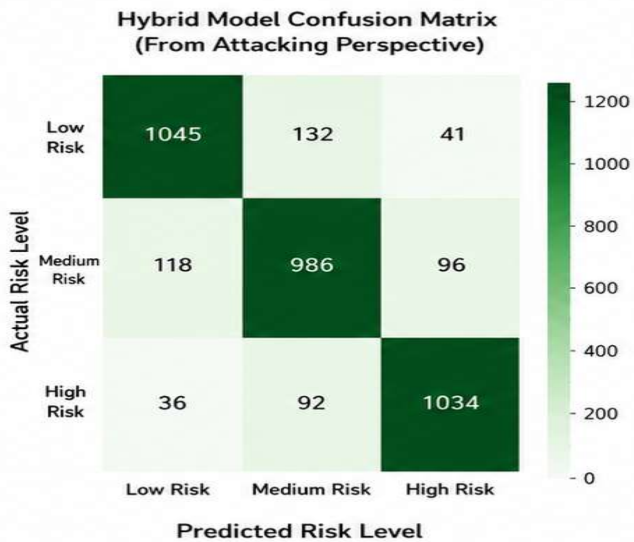


Fig 2: Hybrid Model's Confusion Matrix from an Attacking Perspective, showing model robustness against risky items misclassification.

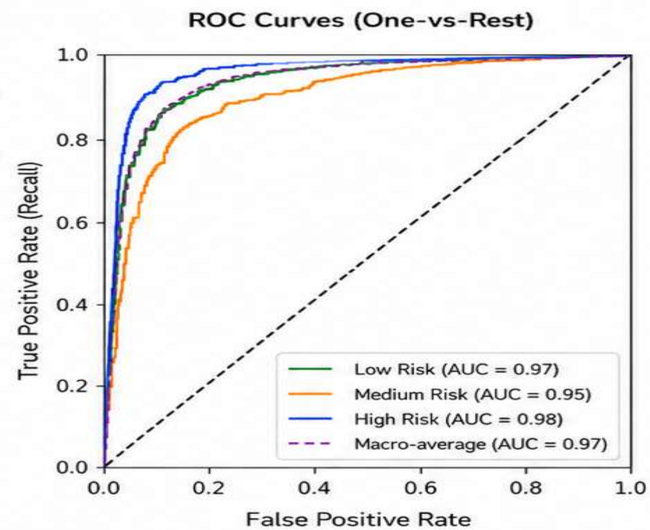


Fig 3: ROC Curves for Risk Levels (One-vs-Rest) demonstrating high AUC scores for the proposed system.

Key Insight: The proposed Perishable Inventory Risk Engine effectively classifies inventory items into Low, Medium, and High Risk with high accuracy and strong generalization, providing reliable support for timely decision-making and risk mitigation.

Conclusion and future work

•**Enhanced Operational Efficiency**: The Perishable Inventory Risk Engine successfully demonstrates that integrating **XGBoost** and **time-series analysis** significantly reduces wastage by providing more accurate expiry predictions than traditional manual methods.

•**Data-Driven Decision Making**: The system transforms raw inventory data into actionable insights, such as dynamic risk scoring and automated alerts, enabling SMEs to maintain profitability while ensuring resource optimization.

•**Sustainability Alignment**: By effectively identifying and managing high-risk items, the project directly supports **SDG Goals 9 and 12**, fostering industrial innovation and promoting responsible, waste-conscious consumption patterns.

References

1. [1] World Bank, Digital Transformation of Small and Medium Enterprises in Developing Economies, World Bank Group Report, 2022.
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5. [5] Rahman, M. and Islam, M., “Smart inventory monitoring system for small businesses using web-based analytics,” International Journal of Information Management Data Insights, vol. 4, no. 2, 2024.

THANK YOU!